Lightweight Speaker Recognition in Poincaré Spaces

Jieun Lee, *Member, IEEE*, Kim Sung-Bin, *Student Member, IEEE*, Seokhyeong Kang, and Tae-Hyun Oh, *Member, IEEE*

***Abstract*—This letter proposes a lightweight model for speaker recognition by leveraging a hyperbolic space. The speaker recognition performance heavily depends on the distinctiveness of speaker embeddings induced by metric learning. However, most state-of-the-art embedding methods are typically based on the Euclidean metric space, which does not account for inherent hierarchical structures of speech voice characteristics. The recent development of the neural hyperbolic geometry has demonstrated its effectiveness to model continuous hierarchical structures, which have been typically cumbersome to model by standard deep neural networks. This facet provides an additional by-product of a compact representation. Inspired by the favorable geometry of the hyperbolic geometry, we developed a hyperbolic ResNet for speaker recognition. We found that in smaller dimension regimes than typical cases, the learned speaker embeddings are more discriminative; in other words, more compact at the same level of performance. Our experiments on the large-scale VoxCeleb datasets show that, given the limited channel dimensions of neural networks, our method consistently has favorable performance against the standard ResNet for both speaker recognition and verification tasks.**

***Index Terms*—Hyperbolic space, speaker recognition, model compression, Poincaré ball model, speaker verification.**

I. Introduction

SPEAKER recognition has diverse practical applications such as speaker diarization, speech recognition, cloud computing security [1], data privacy [2], and video security [3]. The main objective of the speaker recognition task is to determine the identity of a speaker from a speech audio signal. Speaker recognition can be categorized into two common recognition tasks: speaker identification and speaker verification. In all these subtasks, it is favorable to map voices to a discriminative embedding space, where speech voices from different speakers have distinctive embedding vectors in terms of a given metric on the space, while those of the same speaker are embedded nearby.

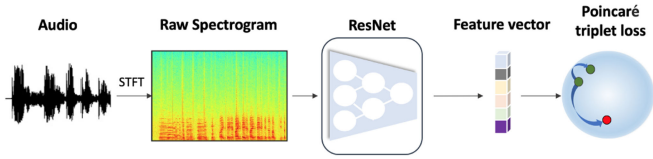


Fig. 1. Illustration of the pipeline of our speaker recognition. Given an utterance of a speaker, a raw spectrogram is first extracted using short-time Fourier transform (STFT). We feed the spectrogram input to ResNet to produce a discriminative embedding which is an utterance-level deep representation. Then, the feature vector is transmitted to the triplet loss function. Finally, the distance from the anchor input to the positive input is minimized, and the distance from the anchor input to the negative input is maximized in the hyperbolic space.

Recently, the end-to-end deep neural networks (DNN) based speaker recognition models [4]–[6] have demonstrated strong performance even with a short utterance and in the unconstrained noisy environment. When it comes to the text-independent speaker recognition problem, to have such robustness, it is required that the DNN models learn an embedding space invariant to language contexts while preserving identifiable voice characteristics, *e.g.*, [7], [8]. Specifically, a standard way of inducing such an embedding space is to take a metric learning approach, where a metric loss induces embedding features to have a small distance within an intra-class (*i.e.*, the same identity) but a large distance between inter-class samples (*i.e.*, different speakers). This can be approached by diverse loss functions [9], such as the contrastive, triplet, and prototypical losses. These losses are measured in a simple Euclidean space or a unit hypersphere. Against this, we raise a question of whether the Euclidean metric space is the best effective and compact way to embed voices.

In this work, we argue that dealing with each dimension of embedding space isotropically does not sufficiently encode discriminative information of input. We focus on inherent hierarchical structures of voice characteristics; *e.g.*, there are similar individuals who have similar voice tones and the way to speak, so that even humans are easily confused. This implies that the main characteristics of similar individuals are easily clustered. On the other hand, while the anatomical structure of human vocal organs is mainly deterministic for an individual, the personal voice may change with respect to one's tone, pitch, and emotional reasons [10]. We postulate that if these hierarchical properties are exploited in the embedding space, we can achieve a compact yet discriminative embedding.

In this letter, we propose a hyperbolic deep neural speaker embedding model that maps an utterance to a compact embedding on a hyperbolic space. Nickel *et al.* [11] demonstrated that

Manuscript received September 11, 2021; revised November 4, 2021; accepted November 9, 2021. Date of publication November 22, 2021; date of current version January 26, 2022. This work was supported by the Institute of Information and Communications Technology Planning and Evaluation (IITP) Grant funded by the Korea government (MSIT) under Grant 2021-0-02068, Artificial Intelligence Innovation Hub under Grant 2019-0-01906, Artificial Intelligence Graduate School Program (POSTECH). The GPU resource was supported by a study on the HPC Support Project, supported by MSIT and NIPA. The associate editor coordinating the review of this article and approving it for publication was Dr. Sri Rama Murty Kodukula. *(Jieun Lee and Kim Sung-Bin contributed equally to this work.) (Corresponding author: T.-H. Oh.)*

Jieun Lee is with the Department of Electrical Engineering, Pohang University of Science and Technology, Pohang 37673, South Korea. She is now with the Korea Education and Research Information Service (KERIS), Daegu 41061, South Korea.

Kim Sung-Bin and Seokhyeong Kang are with the Department of Electrical Engineering, Pohang University of Science and Technology, Pohang 37673, South Korea.

Tae-Hyun Oh is with the Department of Electrical Engineering and Graduate School of Artificial Intelligence, Pohang University of Science and Technology, Pohang 37673, South Korea (e-mail: taehyun@postech.ac.kr).

Digital Object Identifier 10.1109/LSP.2021.3129695

1070-9908 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: National Cheng Chi University. Downloaded on October 18,2026 at 16:24:17 UTC from IEEE Xplore. Restrictions apply.

a continuous tree (hierarchical) structure of data is emerged by imposing hyperbolicity to the embedding space, *i.e.*, Poincaré space. In addition, such hyperbolic embeddings have shown promising empirical results even with much smaller dimensions than that of Euclidean space in natural language processing [12]–[15]. Motivated by this observation, we deploy a hyperbolic geometry, which is defined as a loss function for training the speaker recognition model.

Exploiting the compact hyperbolic embedding space might further contribute to designing the lightweight network architecture. Various network compressions have been consistently developed to reduce computational complexity and memory consumption in typical neural networks. Based on the insights from the parameters or connection redundancy, one approach is the sparsification of large-scale pre-trained neural networks by pruning [16]–[18], quantization [19], and decomposition [20]–[22]. Knowledge distillation [23] can be used as transferring knowledge, such as the representation space and the decision boundary of the original large model (teacher) to the light network. Another line of work is to re-design the network structure in the first place to be lightweight, *e.g.*, slightly modifying the convolution computations as proposed in [24]–[26]. Our method is categorized in this class. We use thin ResNet-34 [5], which is a compressed form of the standard ResNet [27], as our backbone network. By introducing hyperbolicity to the embedding space as aforementioned, we show it can further reduce the size of the model and promising speaker recognition results on even smaller output dimensions.

II. The Poincaré Ball Model of Hyperbolic Space

Before introducing our method, we first review the Poincaré space. The Poincaré space is a hyperbolic space, where hyperbolic n-space $\mathbb{H}^n$ stands for n-dimensional Riemannian manifold with a constant negative sectional curvature, and curvature is measured how a geometric object deviates from a flat plane [28]. We use a *negative* curvature to make the embedding space hyperbolic[1], and use a bounded domain. With these choices, interestingly, as any of two points gets close to the boundary, those distances grow exponentially. This property induced by the hyperbolic geometry spontaneously encourages to embed discriminative hierarchies.

Hyperbolic spaces [14] have been employed due to their highly compact capacity to represent continuous tree-like hierarchical structures (See Fig. 2). Embedding distances induced by the tree-like structures would help express the speaker voice's distinct characteristics. To reflect this hierarchy in the learned embedding space, we integrate hyperbolic space into a speaker recognition framework.

There are five representative models for hyperbolic space, but we choose the Poincaré ball model for the following reasons. Firstly, geodesics and exponential map are derived by a closed-form. Moreover, we note that the Poincaré ball model is appropriate for gradient-based optimization since the Poincaré ball is a conformal model of hyperbolic space [11]. It means that the angles between adjacent vectors are identical to their angles in the Euclidean space. By the convention, the Poincaré ball model

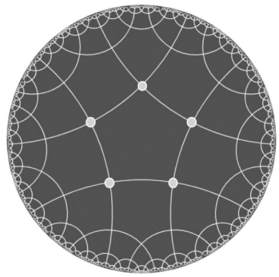


Fig. 2. Visualization of two-dimensional hyperbolic plane (Poincaré ball)[2].

$(\mathbb{D}^n, g^{\mathbb{D}})$ is defined by the manifold $\mathbb{D}^n = \{x \in \mathbb{R}^n : \|x\|_2 < 1\}$ equipped with the Riemannian metric $g_x^{\mathbb{D}} = \lambda_x^2 g^E$, where $\lambda_x := \frac{2}{1-\|x\|_2^2}$, and $g^E = \mathbf{I}_n$ is the Euclidean metric tensor. The induced distance between two points $x, y \in \mathbb{D}^2$ built from a Riemannian metric is given by

$$d_{\mathbb{D}}(x,y) = \cosh^{-1}\left(1 + 2\frac{\|x-y\|_2^2}{\left(1-\|x\|_2^2\right)\left(1-\|y\|_2^2\right)}\right). \quad (1)$$

The distance implies that, as a point gets close to the boundary ($\|x\|_2 \to 1$), the denominator diminishes, and the distance becomes very sensitive to even a small difference. This property is helpful to model personal traits in that an embedding of a very distinctive voice can be placed near a boundary, whereby the voice can be easily distinguished from any other voice. On the other hand, ordinary voices should be carefully compared accordingly, so that identities are correctly distinguished; thus, such voices are embedded far from the boundaries and compared in a similar way to common metric spaces. This can be regarded as adaptive weighting according to regions of the metric space, and the weighting allows to assign different importance according to embedding. This also enables efficient dimension usage.

III. Proposed Method

We introduce a compact speaker recognition neural network model, which embeds a speech utterance onto a learned hyperbolic embedding space. Fig. 1 illustrates the whole pipeline of our proposed method.

A. Backbone Model

We use thin ResNet-34 [5] (Table I) as our main backbone network. In recent years, ResNet [27] have exhibited excellent performance in the field of speaker recognition and speech emotion recognition. Thin ResNet-34 has the same architecture as the original ResNet with 34 layers but uses only one-quarter of the channels like EfficientNet [29] in each residual block to reduce computational cost. To teach the variations between different speakers, mean and variance normalization is applied by instance normalization [30] at the first layer of the model. Additionally, the self-attentive pooling (SAP) [31] is applied to give more attention to informative frame-level features when utterance-level representation is aggregated. The overall network structure is described in Fig. 3.

[1]The Euclidean metric space has *zero* curvature.

[2][Online]. Available: http://www.malinc.se/m/ImageTiling.php

Authorized licensed use limited to: National Cheng Chi University. Downloaded on October 18,2026 at 16:24:17 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ARCHITECTURE OF THIN RESNET-34. THE DIMENSION OF THE OUTPUT VECTOR IS 512, AND THE CHANNEL DIMENSIONS OF THE CONV2-5 ARE 16, 32, 64, AND 128, RESPECTIVELY

Stage	Thin ResNet-34
conv1	7×7, 16, st. 2 3×3, max pool, st. 2
conv2	$\begin{bmatrix} 3 \times 3, & 16 \\ 3 \times 3, & 16 \end{bmatrix} \times 3$, st. 1
conv3	$\begin{bmatrix} 3 \times 3, & 32 \\ 3 \times 3, & 32 \end{bmatrix} \times 4$, st. 2
conv4	$\begin{bmatrix} 3 \times 3, & 64 \\ 3 \times 3, & 64 \end{bmatrix} \times 6$, st. 2
conv5	$\begin{bmatrix} 3 \times 3, & 128 \\ 3 \times 3, & 128 \end{bmatrix} \times 3$, st. 2
FC	9×1, 512, st. 1

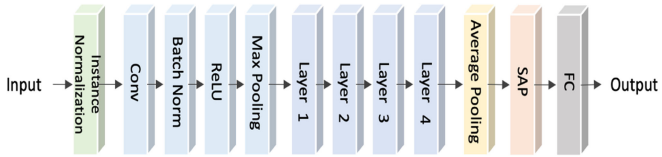


Fig. 3. Illustration of the thin ResNet-34 architecture for speaker recognition. Self-attentive pooling (SAP) [31] is used to give more focus on informative frame-level features.

B. Poincaré Triplet Metric Learning

We train thin ResNet-34 with the Poincaré triplet loss to learn to embed an utterance into hyperbolic space. Since even the same speaker's personalized features can be altered by the various styles of speaking, tone of speech, and the recording environment, we derive a neural embedding by mapping the different utterances of the same speaker as coherently as possible. On the other hand, we encourage to map the embedding farther from the other speakers more distinctively by using the inherent property of the hyperbolic space's non-isometric geometry, where hierarchical structures between embeddings are naturally favored.

We can replace Euclidean space with hyperbolic space because it can be derived from the formalism of Möbius gyrovector spaces in the Poincaré ball model as suggested in hyperbolic neural networks (HNNs) [14]. According to HNNs, the Poincare ball with radius c is defined in the following manner: $\mathbb{D}_c^n = \{x \in \mathbb{R}^n : c\|x\|_2^2 < 1, c \geq 0\}$, where the hyperparameter c controls the curvature of the hyperbolic space. In order to measure the distance between two embeddings, Euclidean vectors should be mapped to general hyperbolic space $\mathbb{D}_c^n$. When c is greater than 0, the induced distance between the two points $x, y \in \mathbb{D}_c^n$ is given as,

$$d_c(x, y) := \frac{2}{\sqrt{c}} \tanh^{-1}\left(\sqrt{c}\,\|-x \oplus_c y\|_2\right), \tag{2}$$

where $\oplus_c$ is the Möbius addition, defined as

$$x \oplus_c y := \frac{(1 + 2c\langle x, y\rangle + c\,\|y\|^2)x + \left(1 - c\,\|x\|^2\right) y}{(1 + 2c\langle x, y\rangle + c^2\,\|x\|^2\,\|y\|^2)}. \tag{3}$$

We can recover Euclidean geometry when $\lim_{c\to 0} d_c(x, y) = 2\|x - y\|_2$. Otherwise, for c=1, the induced distance function on $(\mathbb{D}_c^n, g^c)$ converges to (1); *i.e.*, a general distance that interpolates between Euclidean and Poincaré.

Upon this fact, we propose a Poincaré triplet objective function to train the speaker recognition model. Let each mini-batch contains U utterances from each of S different speakers, whose embeddings are $x_{j,i}$ where $1 \leq j \leq S$ and $1 \leq i \leq U$. Our proposed triplet loss minimizes the distance in hyperbolic space between an anchor and a positive (same identities), and maximizes the distance between an anchor and a negative (different identities) with a margin m as follows:

$$L = \frac{1}{S}\sum_{j=1}^{S} \max(0, d_c(x_{j,0} - x_{j,1}) - d_c(x_{j,0} - x_{k\neq j,1}) + m). \tag{4}$$

In our experiments, we sample the negative utterances from a different speaker within the mini-batch. The negative sample is then chosen by the hard negative mining [31]. Through this letter, we call the thin ResNet-34 architecture trained with the proposed Poincaré triplet loss, as thin HypResNet-34.

IV. EXPERIMENT

In this section, we describe the experimental setup and then evaluate the performance of our proposed method.

A. Experiment Setup

1) Dataset: We use the VoxCeleb2 [32] dataset[3], which contains over a million utterances obtained from over 6,000 speakers for the training phase. The VoxCeleb1 [33] and DR-VCTK[4] benchmark datasets are used for evaluating the test performance of speaker verification and identification.

2) Acoustic features: From each utterance of the recorded voices, we randomly extract a fixed-length of temporal segments during training. Then we convert the audio signal into frequency domains by constructing the 257-dimensional raw spectrograms as acoustic features. The raw spectrograms are computed with short frames by Hamming windowing of length 25 ms and of shift 10 ms.

3) Training: All the experiments in this letter are implemented by PyTorch framework [34]. We set the batch size as 350 and train neural networks on NVIDIA Tesla V100 GPUs. The Adam optimizer with the initial learning rate of 0.001, decreasing by 5% every 10 epochs, was used. We tune the curvature $c \in \{0.3, 0.03, 0.003\}$ separately for each run. The output dimension and the number of channels in each layer are decreased by the same ratio; *i.e.*, uniformly reduce the channel size by the same factor. For example, if the output dimension is reduced from 512 to 64 by 0.125 times, the number of channels in the `conv3-5` stages becomes $[4, 8, 16]$, respectively.

B. Evaluation

In the following, we evaluate the performance of the deep speaker hyperbolic embedding by comparing it with its Euclidean counterpart in small output dimension regimes so that

[3]http://www.robots.ox.ac.uk/~vgg/data/voxceleb/
[4]https://datashare.is.ed.ac.uk/handle/10283/3038

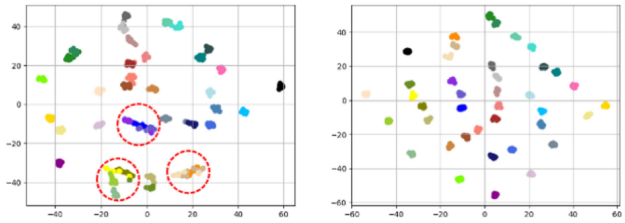
Authorized licensed use limited to: National Cheng Chi University. Downloaded on October 18,2026 at 16:24:17 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SPEAKER VERIFICATION TEST ON THE VOXCELEB1 DATASET. WE RUN THE ALGORITHMS UNTIL 100 EPOCHS, AND THE BOLD NUMBERS DENOTE THE BEST PERFORMANCE FOR EACH COLUMN. IN FACT, THE THIN RESNET-34 MODEL HAS ZERO CURVATURE

Output dimension	Model	Curvature	Loss function	EER (%)	Params
64	Thin ResNet-34	-	triplet	11.59	0.72M
	Thin HypResNet-34	0.3	Poincare triplet	37.54	
		0.03		**10.96**	
		0.003		11.44	
32	Thin ResNet-34	-	triplet	19.78	0.67M
	Thin HypResNet-34	0.3	Poincare triplet	13.65	
		0.03		13.98	
		0.003		**13.56**	

TABLE III
ACCURACY OF SPEAKER IDENTIFICATION ON VOXCELEB1 AND DR-VCTK. BOLD NUMBERS DENOTE THE BEST PERFORMANCE IN EACH ROW

Dataset	Output Dimension	Thin ResNet-34	Thin HypResNet-34
VoxCeleb1	64	88.4	**88.6**
	32	82.2	**84.9**
	24	79.4	**80.0**
	16	74.8	**78.2**
DR-VCTK	64	79.8	**82.6**
	32	52.3	**67.0**



(a) Thin ResNet-34 (b) Thin HypResNet-34

Fig. 4. t-SNE visualization [35] of learnt speaker embeddings. The plots show two-dimensional t-SNE embeddings obtained from the thin ResNet-34 model (a) and the thin HypResNet-34 model (b), respectively, which output 32-dimensional vectors. We color the embeddings using the speaker labels.

the benefits of hyperbolic embedding in small dimensions are assessed. The training method with hyperbolic properties contributes to improving the performance in the lightweight model regimes. A breakdown of this is given in Tables II and III for speaker verification and identification, respectively.

Table II demonstrates the comparison of the speaker verification performance between the original thin ResNet-34 model trained with the triplet loss on Euclidean embedding space and the thin HypResNet-34 model. Given the same embedding dimension, thin HypResNet-34 shows improvement over the original thin ResNet-34 even when the overall output dimension and the size of the model parameters are dramatically diminished compared to the original one. We hypothesize that the improvement is obtained by the hyperbolic embedding space, which encourages distinctive speaker representation by its embedding geometry. This improvement is consistently shown in the speaker identification task (See Table III). We trained the models with early stopping. We observe that up to the 16-dimensions, our proposed method outperforms the baseline model, and further decreasing the output size leads to unstable training.

The learned embedding quality is qualitatively shown in Fig. 4. The figure shows the comparison between the learned speaker embeddings of the baseline and the proposed model, using the t-SNE visualization [35]. As shown, it is clear that our method produces well-separated clusters even with the small output dimensions, while the baseline exhibits unclear or noisy clusters.

Lastly, in Table IV, we observe that the proposed model performs favorably against the baseline model even in a short utterance length. We conclude that exploiting hyperbolic space enables compact speaker embedding and performs favorably against the baseline model in the regimes of smaller models and smaller output dimensions.

TABLE IV
PERFORMANCE COMPARISON ON THE SHRINKAGE OF THE AUDIO SIGNALS. TEST SAMPLES FROM VOXCELEB1 ARE MOSTLY COMPRISED OF 5-6 SECONDS, WHICH IS TERMED BY T

Query Length	Output Dimension	Model	EER(%)
$\frac{T}{2}$	64	Thin ResNet-34	14.67
		Thin HypResNet-34	**14.49**
	32	Thin ResNet-34	22.41
		Thin HypResNet-34	**17.49**

V. DISCUSSION

In this letter, we have introduced a thin HypResNet-34 model that can leverage the hyperbolic property into speaker recognition tasks. The model surpasses the performance of the previous work on lightweight model parameter conditions. Specifically, we exploit the multi-scale nature of information in the sequential speech signals by implicitly imposing hyperbolic geometry. Our compact and efficient speaker model, thin HypResNet-34, may broaden interesting applications, *e.g.*, multi-modal person verification [36], [37] to be efficient, speaker face prediction from voice identity [38] to be on-device, and speech separation by voice identity [39]. Investigating those application scenarios would be interesting future directions.

Authorized licensed use limited to: National Cheng Chi University. Downloaded on October 18,2026 at 16:24:17 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] J. Li, J. Li, X. Chen, C. Jia, and W. Lou, "Identity-based encryption with outsourced revocation in cloud computing," *IEEE Trans. Comput.*, vol. 64, no. 2, pp. 425–437, Feb. 2013.

[2] P. Li, J. Li, Z. Huang, C.-Z. Gao, W.-B. Chen, and K. Chen, "Privacy-preserving outsourced classification in cloud computing," *Cluster Comput.*, vol. 21, no. 1, pp. 277–286, 2018.

[3] Z. Pan, J. Lei, Y. Zhang, X. Sun, and S. Kwong, "Fast motion estimation based on content property for low-complexity h. 265/hevc encoder," *IEEE Trans. Broadcast.*, vol. 62, no. 3, pp. 675–684, Sep. 2016.

[4] K. Okabe, T. Koshinaka, and K. Shinoda, "Attentive statistics pooling for deep speaker embedding," 2018, *arXiv:1803.10963*.

[5] W. Xie, A. Nagrani, J. S. Chung, and A. Zisserman, "Utterance-level aggregation for speaker recognition in the wild," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2019, pp. 5791–5795.

[6] C. Li *et al.*, "Deep speaker: An end-to-end neural speaker embedding system," 2017, *arXiv:1705.02304*.

[7] M. Tagliasacchi, B. Gfeller, F. de Chaumont Quitry, and D. Roblek, "Pre-training audio representations with self-supervision," *IEEE Signal Process. Lett.*, vol. 27, pp. 600–604, 2020.

[8] R. Travadi and S. Narayanan, "Total variability layer in deep neural network embeddings for speaker verification," *IEEE Signal Process. Lett.*, vol. 26, no. 6, pp. 893–897, Jun. 2019.

[9] L. Wan, Q. Wang, A. Papir, and I. L. Moreno, "Generalized end-to-end loss for speaker verification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2018, pp. 4879–4883.

[10] V. Srinivas and C. Santhirani, "Optimization-based support vector neural network for speaker recognition," *Comput. J.*, vol. 63, no. 1, pp. 151–167, 2020.

[11] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6338–6347.

[12] C. De Sa, A. Gu, C. Ré, and F. Sala, "Representation tradeoffs for hyperbolic embeddings," in *Proc. Mach. Learn. Res.*, vol. 80, 2018, Art. no. 4460.

[13] C. Gulcehre *et al.*, "Hyperbolic attention networks," 2018, *arXiv:1805.09786*.

[14] O. Ganea, G. Bécigneul, and T. Hofmann, "Hyperbolic neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 5345–5355.

[15] A. Tifrea, G. Bécigneul, and O.-E. Ganea, "Poincaré glove: Hyperbolic word embeddings," 2018, *arXiv:1810.06546*.

[16] S. Han, J. Pool, J. Tran, and W. J. Dally, "Learning both weights and connections for efficient neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1135–1143.

[17] H. Li, A. Kadav, I. Durdanovic, H. Samet, and H. P. Graf, "Pruning filters for efficient convnets," 2016, *arXiv:1608.08710*.

[18] E. Tartaglione, S. Lepsøy, A. Fiandrotti, and G. Francini, "Learning sparse neural networks via sensitivity-driven regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 3878–3888.

[19] K. Wang, Z. Liu, Y. Lin, J. Lin, and S. Han, "HAQ: Hardware-aware automated quantization with mixed precision," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2019, pp. 8612–8620.

[20] V. Lebedev, Y. Ganin, M. Rakhuba, I. Oseledets, and V. Lempitsky, "Speeding-up convolutional neural networks using fine-tuned CP-decomposition," in *Proc. Int. Conf. Learn. Representations*, vol. abs/1412.6, 2015, pp. 1–10.

[21] C. Tai, T. Xiao, X. Wang, and E. Weinan, "Convolutional neural networks with low-rank regularization," in *Proc. Int. Conf. Learn. Representations*, 2016.

[22] A.-H. Phan *et al.*, "Stable low-rank tensor decomposition for compression of convolutional neural network," in *Proc. Eur. Conf. Comput. Vision*, 2020, pp. 522–539.

[23] T. Li, J. Li, Z. Liu, and C. Zhang, "Few sample knowledge distillation for efficient network compression," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2020, pp. 14639–14647.

[24] A. G. Howard *et al.*, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[25] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2018, pp. 4510–4520.

[26] N. Hyeon-Woo, M. Ye-Bin, and T.-H. Oh, "FedPara: Low-rank hadamard product parameterization for efficient federated learning," 2021, *arXiv:2108.06098*.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2016, pp. 770–778.

[28] V. Khrulkov, L. Mirvakhabova, E. Ustinova, I. Oseledets, and V. Lempitsky, "Hyperbolic image embeddings," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2020, pp. 6418–6428.

[29] M. Tan and Q. Le EfficientNet, "Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[30] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: The missing ingredient for fast stylization," 2016, *arXiv:1607.08022*.

[31] J. S. Chung *et al.*, "In defence of metric learning for speaker recognition," 2020, *arXiv:2003.11982*.

[32] J. S. Chung, A. Nagrani, and A. Zisserman, "VoxCeleb2: Deep speaker recognition," 2018, *arXiv:1806.05622*.

[33] A. Nagrani, J. S. Chung, and A. Zisserman, "VoxCeleb: A large-scale speaker identification dataset," 2017, *arXiv:1706.08612*.

[34] A. Paszke *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, vol. 32, pp. 8026–8037.

[35] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

[36] S. Shon, T.-H. Oh, and J. Glass, "Noise-tolerant audio-visual online person verification using an attention-based neural network fusion," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2019, pp. 3995–3999.

[37] C. Kim, H. V. Shin, T.-H. Oh, A. Kaspar, M. Elgharib, and W. Matusik, "On learning associations of faces and voices," in *Proc. Asian Conf. Comput. Vision*, 2018, pp. 276–292.

[38] T.-H. Oh *et al.*, "Speech2Face: Learning the face behind a voice," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2019, pp. 7531–7540.

[39] A. Ephrat *et al.*, "Looking to listen at the cocktail party: A speaker-independent audio-visual model for speech separation," *ACM Trans. Graph.*, vol. 37, no. 4, pp. 1–11, 2018.

Authorized licensed use limited to: National Cheng Chi University. Downloaded on October 18,2026 at 16:24:17 UTC from IEEE Xplore. Restrictions apply.